

Dimension Reduction Analysis: pathmnist

Generated report #2

Seed: 0 | Plan revision: 1

Dataset profile & preprocessing

PathMNIST, part of the MedMNIST v2 collection. Derived from a histopathology study of colorectal cancer: 28x28 RGB image patches extracted from hematoxylin and eosin (H&E) stained histology slides, downsampled from an original resolution of 224x224. Source: NCT-CRC-HE-100K (train/validation split) and CRC-VAL-HE-7K (test split, from a different clinical center; not loaded by this project). Only the train split is used.

Samples: 89,996 Features: 2,352 Sparsity: 0.0% Missing values: No

Preprocessing: scale_unit_range

Raw pixel intensities are 0-255 (profile.json value_range) and DATA_DESCRIPTION.md confirms these are bounded RGB values, not counts; rescaling to [0,1] keeps no single pixel channel dominating a distance computation purely from its raw scale. Not count-like data, so normalize_total/log1p don't apply, and n_features (2,352) is well under n_samples (89,996) so select_hvg isn't needed either.

Methods and hyperparameters

pca (role: general_purpose, status: ok)

Hyperparameters: `n_components=50`

Required linear baseline. `n_components = min(50, floor(89996/2), 2352) = 50` per the standard heuristic; plenty of samples relative to components so no overfitting risk here.

sparse_pca (role: general_purpose, status: ok)

Hyperparameters: `n_components=50`

Required PCA variant. `n_samples` (89,996) exceeds the 5,000-sample threshold where `kernel_pca`'s $O(N^2)$ kernel matrix becomes impractical, so `sparse_pca` is used instead, per method-selection/SKILL.md's size-gated rule.

umap (role: visualization_only, status: ok)

Hyperparameters: `n_neighbors=30`, `min_dist=0.1`, `n_components=2`, `random_state=0`

Standard visualization method; `n_neighbors` raised to 30 (from the default 15) because `n_samples` (89,996) exceeds 20,000, per method-selection/SKILL.md's size heuristic, for a smoother, more stable neighborhood graph at this scale.

Quantitative metrics

method	trustworthiness	variance explained (PCs used)	PCs for 90% variance	label silhouette
pca	0.9820	0.7907	178	-0.0664
sparse_pca	0.9816			-0.0487
umap	0.8127			-0.0588

How to read the metrics

Trustworthiness (`sklearn.manifold.trustworthiness`, 10 neighbours)

For each sample, takes its 10 nearest neighbours in the embedding and penalizes any that were not close neighbours in the original preprocessed data, weighting each penalty by how far down the original neighbour ranking that point sat. Ranges 0 to 1; 1 means the embedding introduces no false neighbours. It measures local neighbourhood fidelity only, not preservation of global distances. Computed against the data after the plan's preprocessing steps; on a random 5,000-sample subset when the dataset is larger than that (see Limitations).

Variance explained (PCs used)

The fraction of the total variance in the preprocessed data captured by the principal components actually kept in the embedding: the sum of those components' explained-variance ratios (each component's variance divided by the total). Ranges 0 to 1. PCA only; blank for methods it does not apply to.

PCs for 90% variance

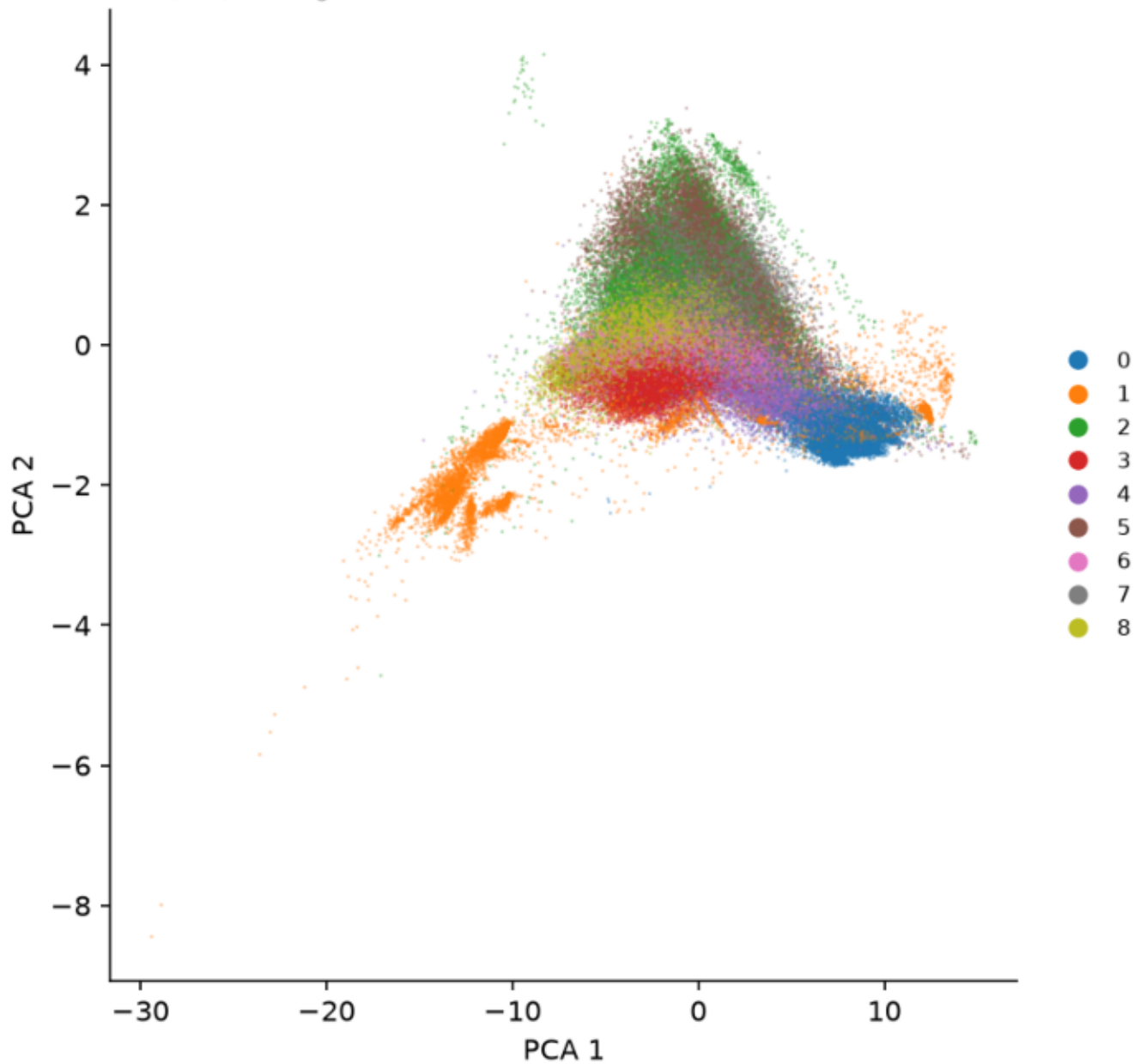
The smallest number of principal components whose cumulative explained variance reaches 90%, read off a diagnostic PCA fit of up to $\min(4 \times n_{\text{components}}, 200)$ components. A leading '>' means 90% was not reached within that fit, so the true number is larger than shown. The scree plots show the full curve.

Label silhouette

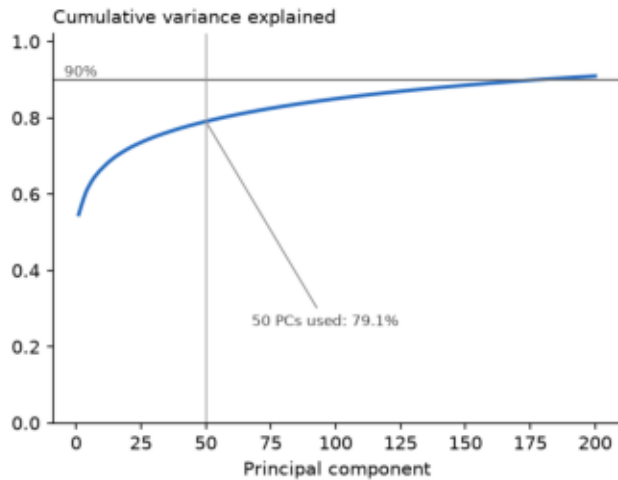
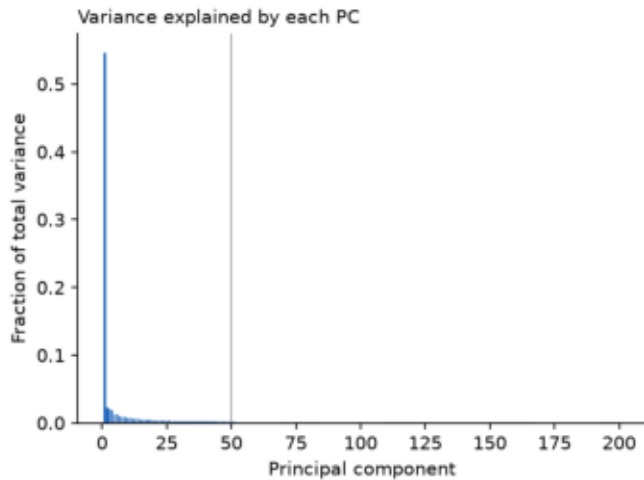
Mean silhouette coefficient (`sklearn.metrics.silhouette_score`) of the dataset's ground-truth class labels in the embedding space, using all embedding dimensions. Ranges -1 to 1: near 1 means points sit close to their own class and far from others, near 0 means classes overlap, negative means points are on average nearer another class than their own. A supervised sanity check only; labels were never used to build an embedding or to tune any hyperparameter.

pathmnist: pca

n=89,996, showing dims 1-2 of 50

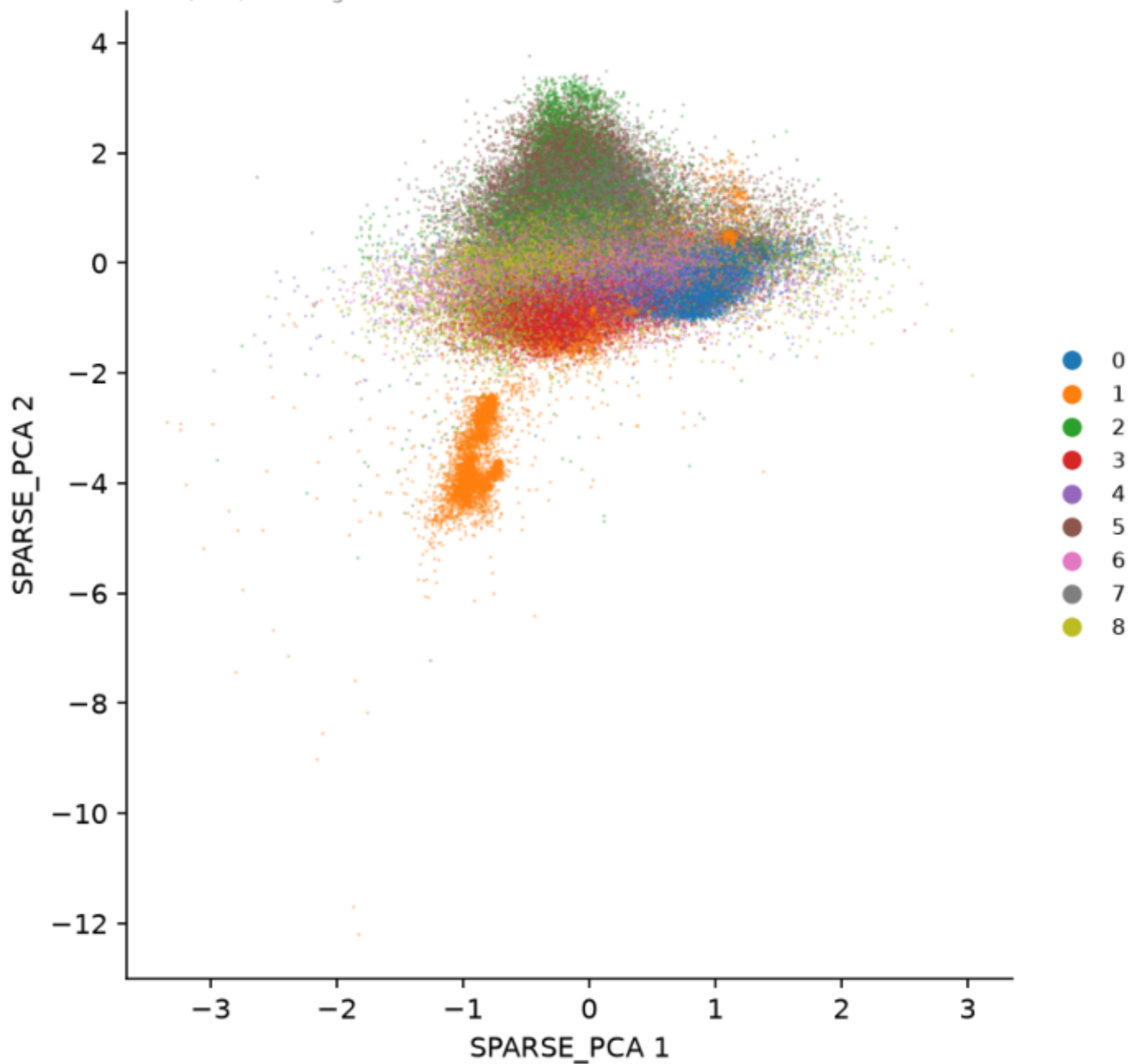


pathmnist: pca scree plot



pathmnist: sparse_pca

n=89,996, showing dims 1-2 of 50



pathmnist: umap

n=89,996



Findings and interpretation

Method results

PCA (general purpose, 50 components) has trustworthiness 0.9820 and explains 0.7907 of the total variance; 90% is reached at 178 components. Sparse PCA (general purpose, 50 components) is nearly identical on trustworthiness at 0.9816. UMAP (visualization only, 2 components) has trustworthiness 0.8127. The UMAP score is not comparable head to head with the two 50-dimensional PCA-family scores, since a 2-dimensional layout is expected to preserve less neighbourhood structure. All three trustworthiness values were computed on a random subset of 5,000 of the 89,996 samples, so treat them as estimates.

Labels

The label silhouette is slightly negative for all three methods (PCA -0.0664, Sparse PCA -0.0487, UMAP -0.0588). By this measure, none of the embeddings groups the nine tissue classes tightly. The plots agree in a limited way: labels 0 and 1 occupy visibly distinct regions in both the PCA and UMAP plots, while the other classes overlap heavily. This means raw 28x28 pixel intensities alone do not separate most of the classes in these embeddings; it does not mean the classes are indistinguishable in general. The labels were used only for coloring and for this sanity check, never to build an embedding or tune a hyperparameter.

Limitations

Trustworthiness was computed on a random subset of 5,000 of 89,996 samples (it is $O(N^2)$ and infeasible at full size) for: `pca`, `sparse_pca`, `umap`. Treat those scores as estimates, not exact full-dataset values.

The label silhouette column is a supervised sanity check only, computed on the embedding's own dimensions; it was never used to tune any hyperparameter.

No adversarial critique/revision pass was performed for this run (critic subagent not yet built).